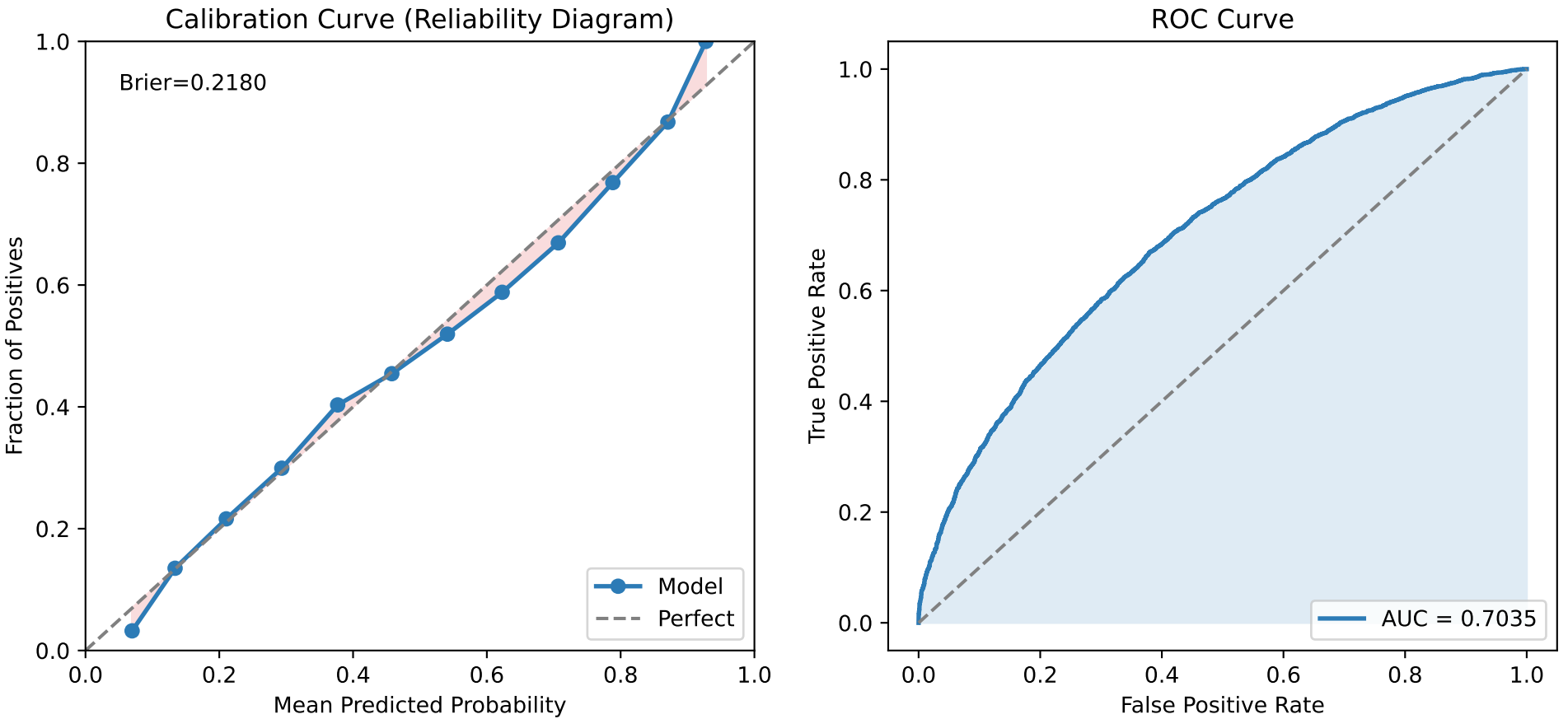
Tennis ML Backtest Report

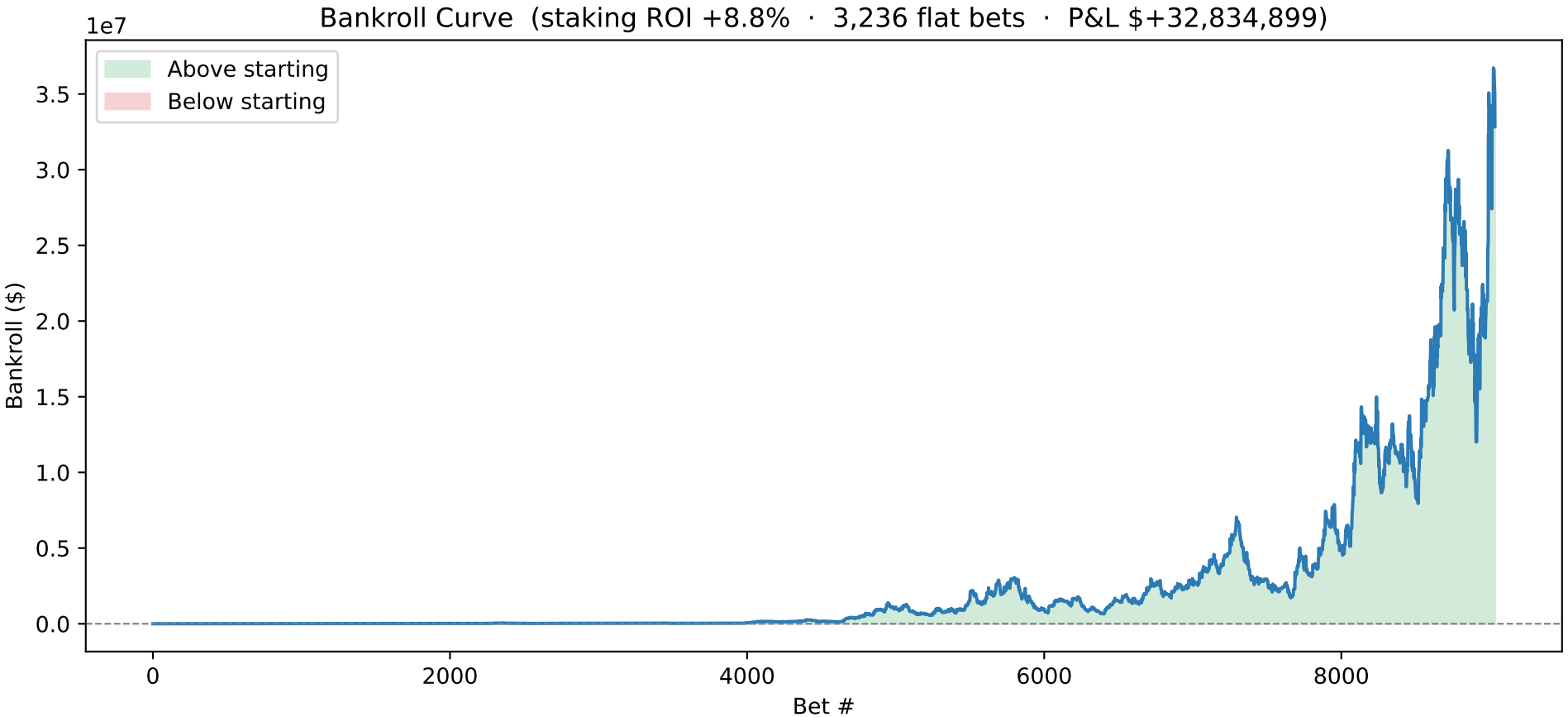
XGBoost · Walk-Forward · Out-of-Sample 2022–2024

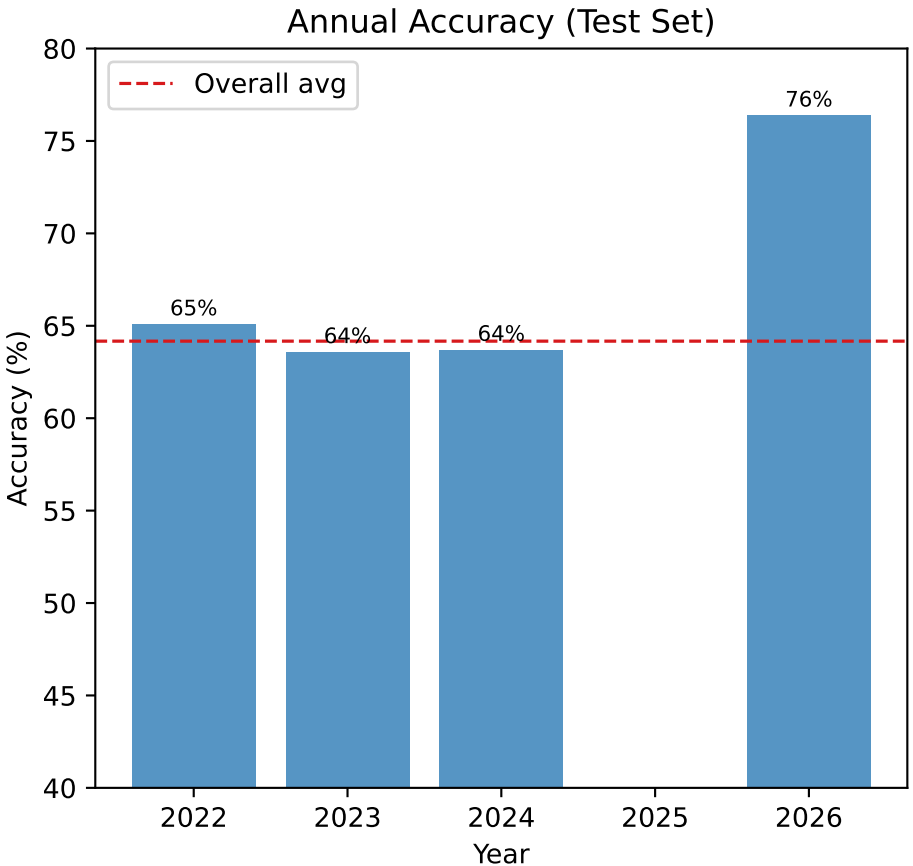
Metric	Value	Note
Matches Evaluated	9,034	test set (2022–2024)
Accuracy	64.17%	higher = better
ROC-AUC	0.7035	1.0 = perfect
Brier Score	0.2180	lower = better
Log-Loss	0.6243	lower = better
Bets Triggered	3,236	$10 flat bet, +4pp edge min
Win Rate (bets)	43.9%	of triggered bets
Staking ROI	+8.8%	profit ÷ total staked (after 8% vig)
Cumulative P&L	$+32,834,899	on $371,708,524 staked

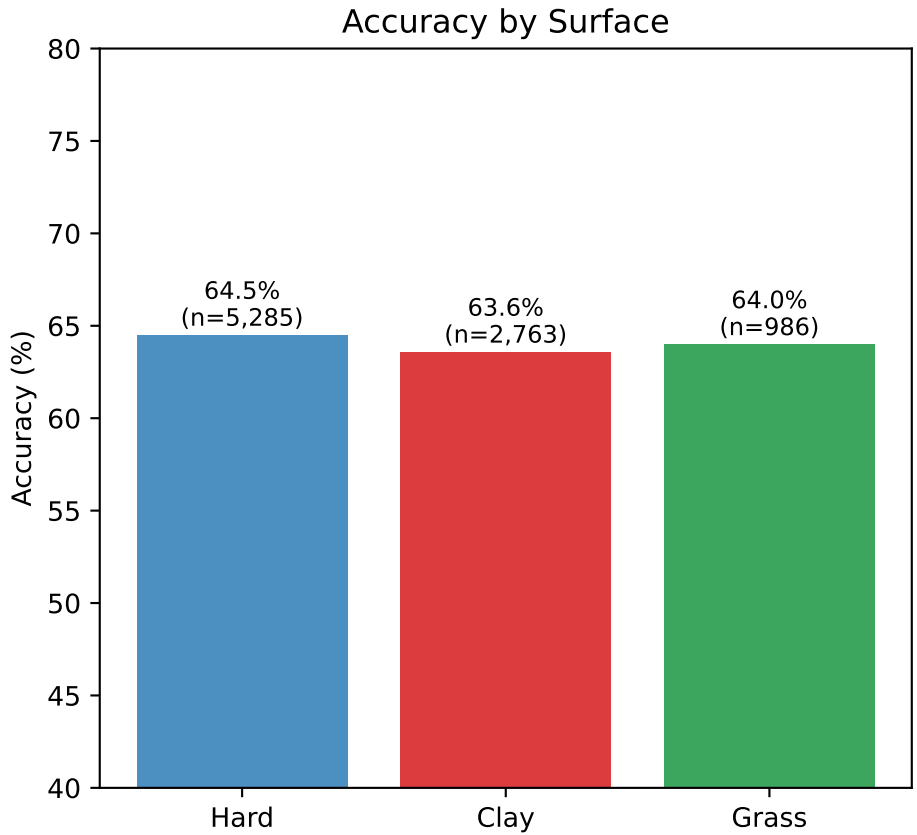
Methodology

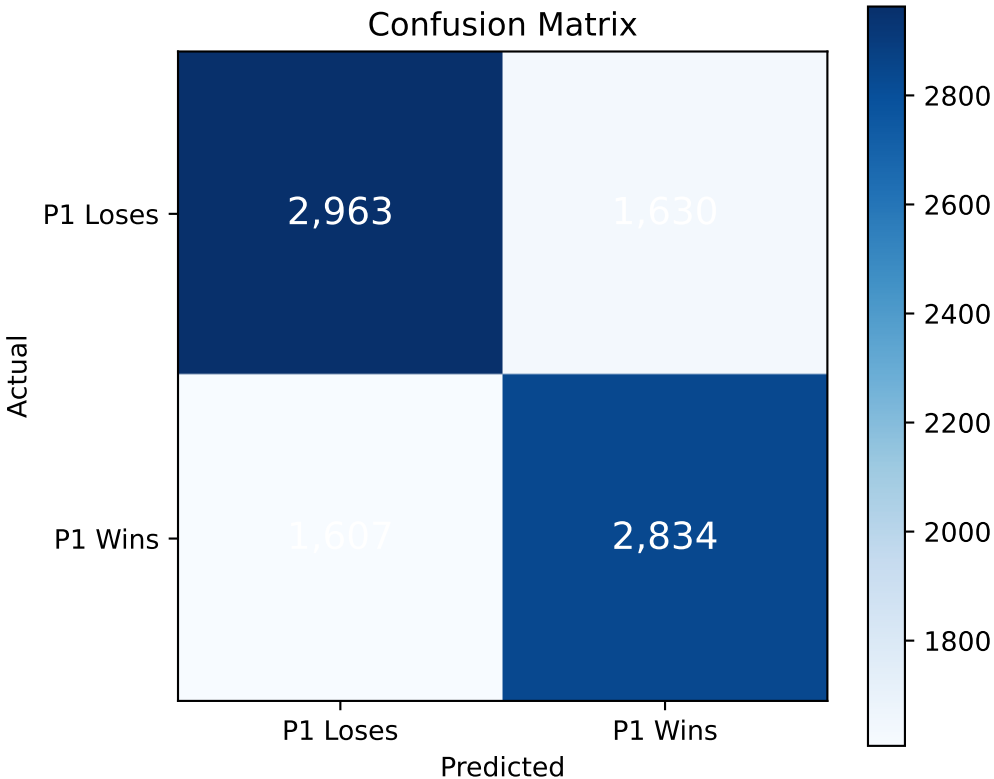
- Data: All ATP main-tour match CSVs (1968–2024) — Jeff Sackmann dataset
- Model: XGBoost trained on 1968–2019; evaluated on 2022–2024 (unseen)
- Features (14): surface dummies, best-of, handedness, height, age, ATP rank, ATP rank-points — all taken from the match row (no lookahead)
- ELO ratings: updated match-by-match; prediction uses pre-match ELO
- Betting sim: 25% fractional Kelly vs. ELO-implied market probability Minimum edge threshold: 4 pp; position cap: 10% of bankroll

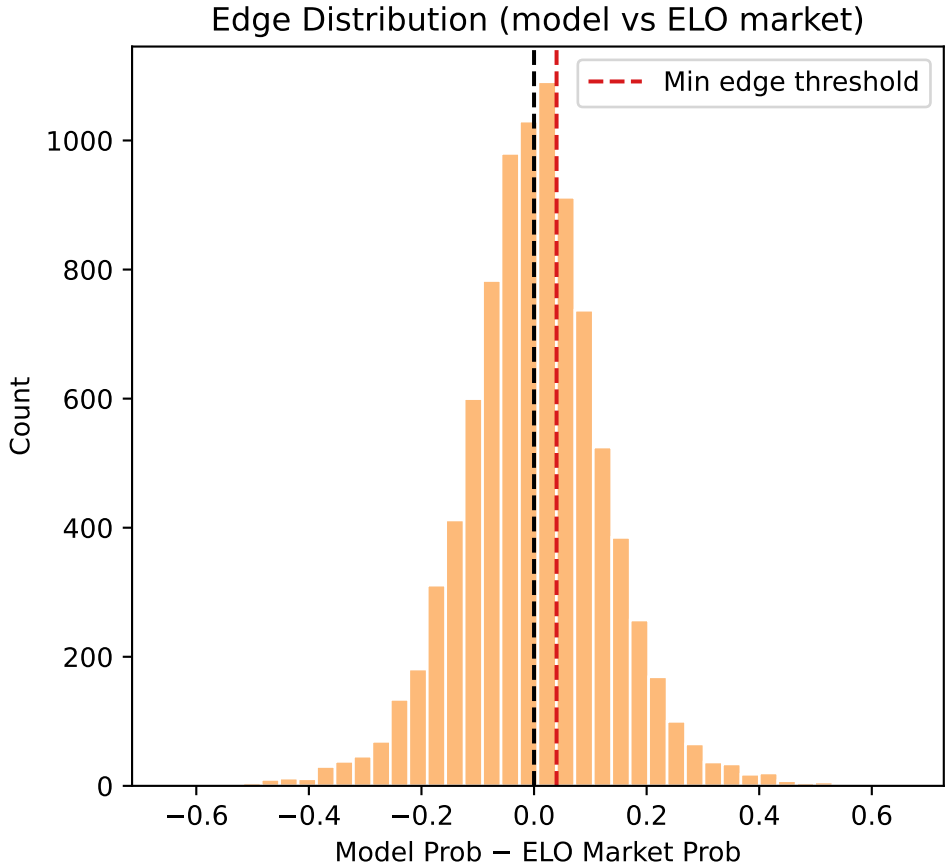

Model Calibration & Discrimination (Test 2022–2024)
Calibration Curve (Reliability Diagram)
Brier=0.2180
Fraction of Positives
Mean Predicted Probability
Model
Perfect
0.0
0.2
0.4
0.6
0.8
1.0
ROC Curve
True Positive Rate
False Positive Rate
AUC = 0.7035

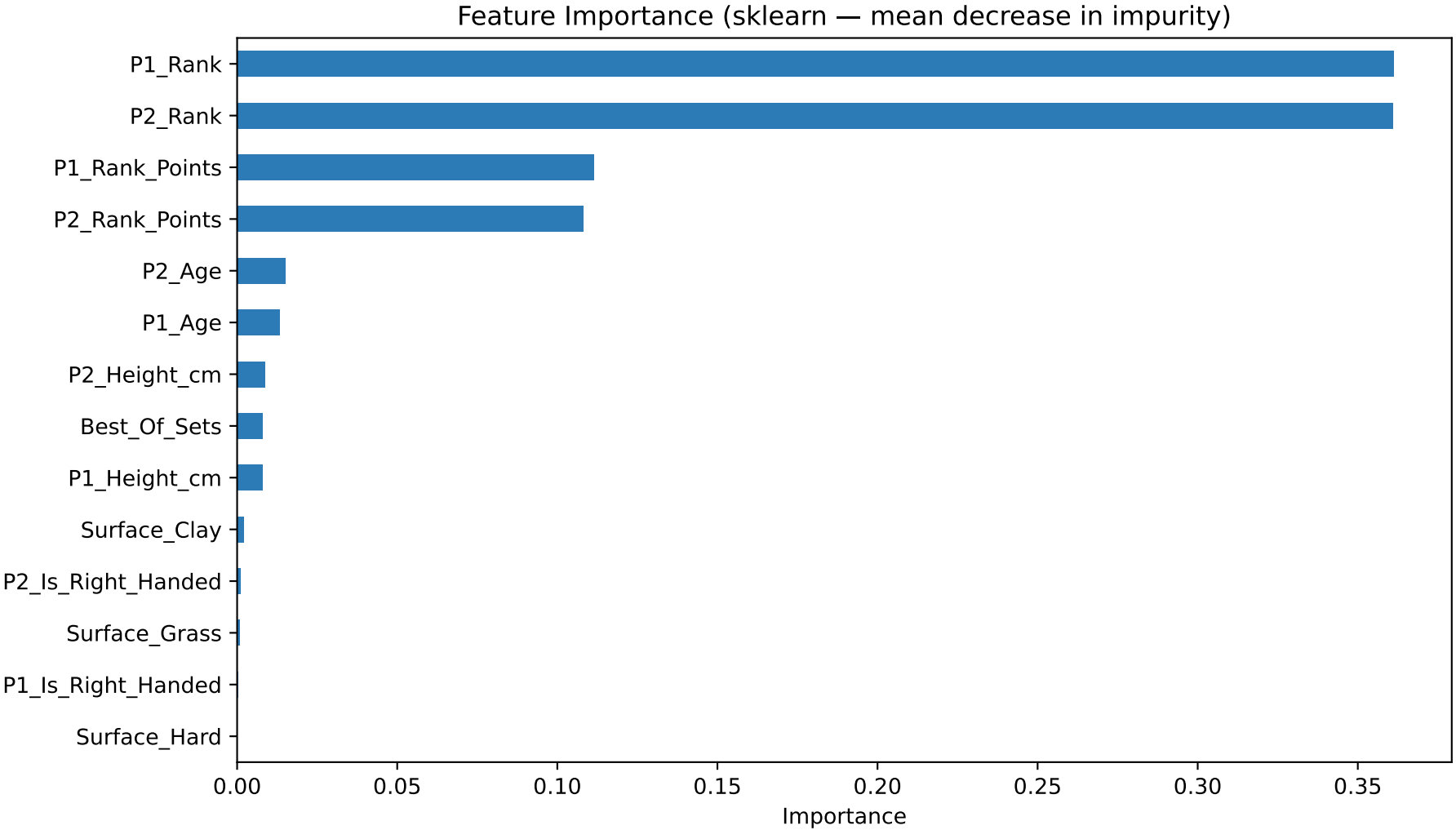

Bankroll Curve (staking ROI +8.8% · 3,236 flat bets · P&L $+32,834,899)
1e7
Above starting
Below starting
Bankroll ($)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
2000
4000
6000
8000
Bet #

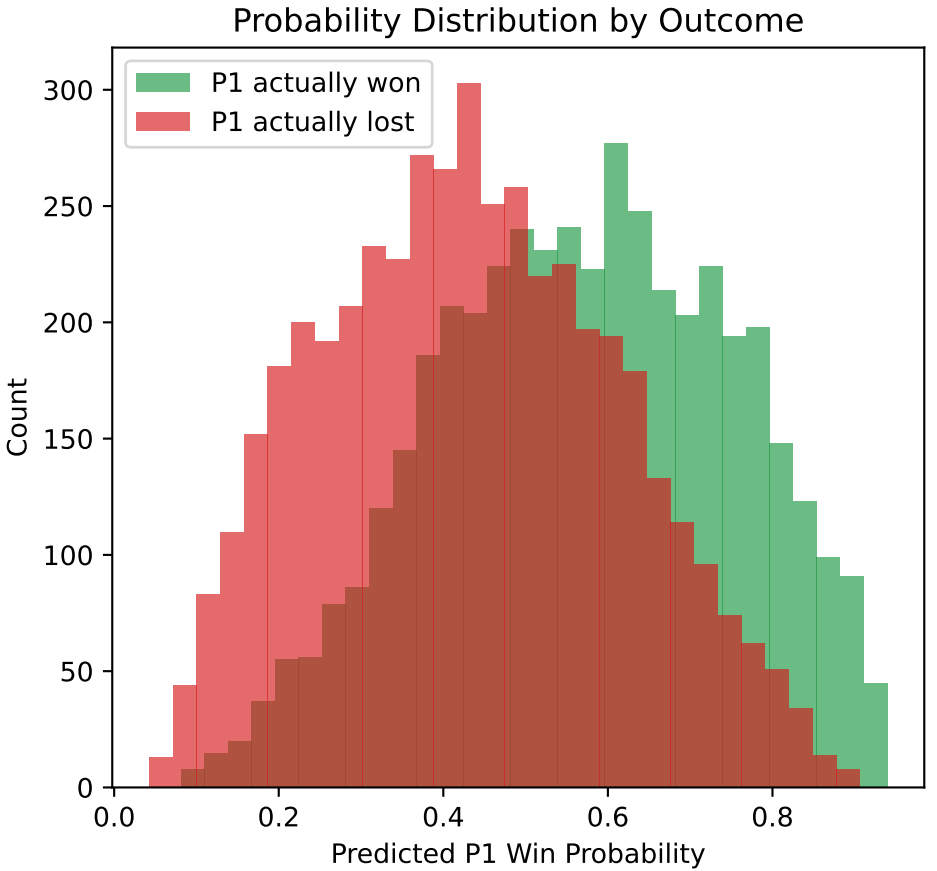

Annual Accuracy (Test Set)
Overall avg
65%
64%
64%
76%
Accuracy (%)
80
75
70
65
60
55
50
45
40
2022
2023
2024
2025
2026
Year

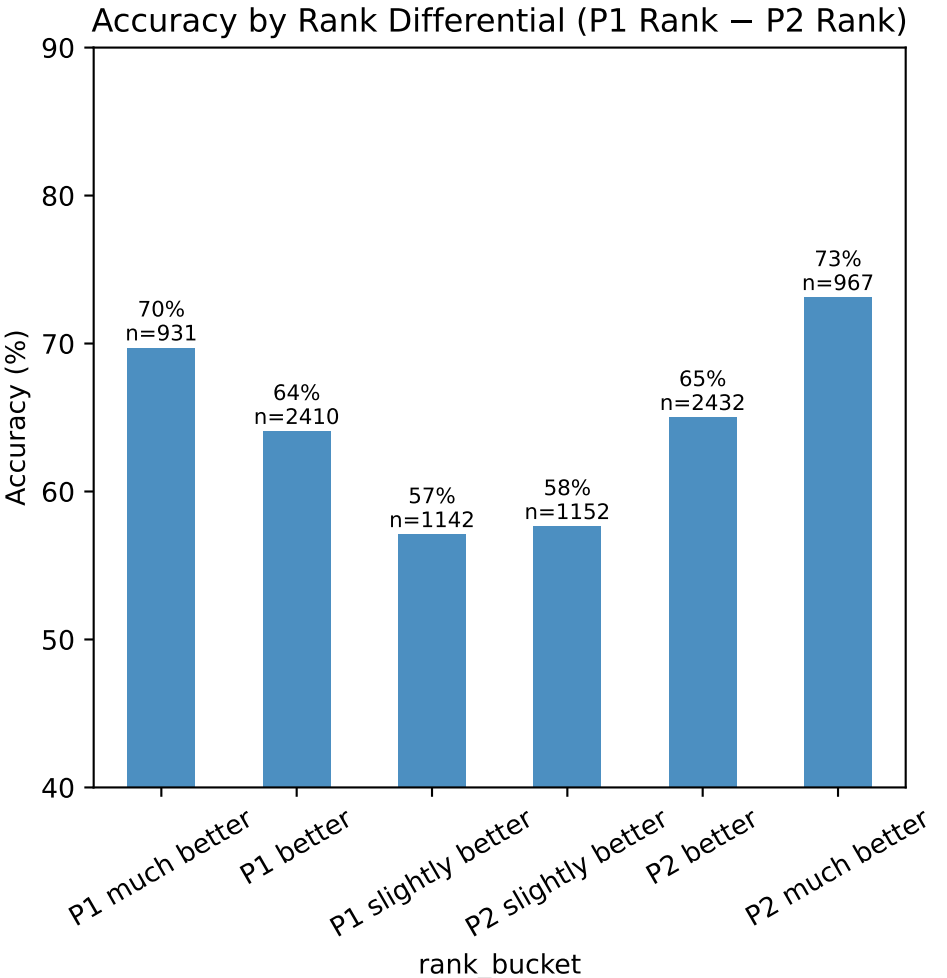

Accuracy by Surface
64.5%
(n=5,285)
63.6%
(n=2,763)
64.0%
(n=986)
Accuracy (%)
80
75
70
65
60
55
50
45
40
Hard
Clay
Grass


Confusion Matrix
P1 Loses
P1 Wins
Actual
2,963
1,630
1,607
2,834
P1 Loses
P1 Wins
Predicted
2800
2600
2400
2200
2000
1800


Edge Distribution (model vs ELO market)
Min edge threshold
Count
1000
800
600
400
200
0
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Model Prob − ELO Market Prob


Feature Importance (sklearn — mean decrease in impurity)
P1_Rank
P2_Rank
P1_Rank_Points
P2_Rank_Points
P2_Age
P1_Age
P2_Height_cm
Best_Of_Sets
P1_Height_cm
Surface_Clay
P2_Is_Right_Handed
Surface_Grass
P1_Is_Right_Handed
Surface_Hard
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Importance


Probability Distribution by Outcome
P1 actually won
P1 actually lost
Count
Predicted P1 Win Probability
0
50
100
150
200
250
300
0.0
0.2
0.4
0.6
0.8


Accuracy by Rank Differential (P1 Rank − P2 Rank)
Accuracy (%)
40
50
60
70
80
90
70%
n=931
64%
n=2410
57%
n=1142
58%
n=1152
65%
n=2432
73%
n=967
P1 much better
P1 better
P1 slightly better
P2 slightly better
P2 better
P2 much better
rank_bucket

Methodology & Anti-Leakage Notes

DATA

- Source: Jeff Sackmann's tennis_atp repository — all main-tour match CSVs from 1968 through 2024 (~180,000+ matches).
- Only atp_matches_YYYY.csv files are used (main tour only).

TEMPORAL SPLITS

- Warm-up / Training reference : 1968–2019
- Validation : 2020–2021 (not reported)
- Out-of-sample Test : 2022–2024 (all metrics here)

ANTI-LEAKAGE MEASURES

1. ELO ratings are updated *after* each match is processed; each row's prediction uses only pre-match ELO.
2. ATP rank/rank-points come from the match row itself — a snapshot of what was known at match time.
3. The XGBoost model was trained on 1968–2019 data only. Test years 2022–2024 were never seen during training.
4. P1/P2 assignment is randomised with a fixed seed (42) so the model cannot learn from label asymmetry.

BETTING SIMULATION

- Market implied probability: ELO-based (pre-match ELO difference converted to win probability via the logistic formula).
- Bet triggered when: model_prob − market_prob > 4 pp (post-vig).
- Sizing: flat $10 per bet — avoids compounding distortions.
- 8% vig applied to simulate Kalshi exchange rake / spread.
- ROI = cumulative profit ÷ total amount staked (not bankroll).
- Note: ELO is less efficient than real markets; edge vs real Kalshi prices will be smaller. Results are a theoretical bound.

FEATURES (14)

Surface_Hard, Surface_Clay, Surface_Grass, Best_Of_Sets, P1_Is_Right_Handed, P1_Height_cm, P1_Age, P1_Rank, P1_Rank_Points, P2_Is_Right_Handed, P2_Height_cm, P2_Age, P2_Rank, P2_Rank_Points

LIMITATIONS

- ELO does not account for surface, fitness, or head-to-head.
- No transaction costs, slippage, or market liquidity modelled.
- Real-world Kalshi market prices will differ from ELO-implied.